

## 2017-06-28 TRAFFIC ANALYSIS EXERCISE - ANSWERS

---

### QUESTIONS:

- What is the MAC address, IP address, and host name of the infected Windows computer?
- What is the date and time (in UTC) the computer was infected?
- Based on the Snort and Suricata alerts, what was the computer infected with?
- Based on indicators from first HTTP GET request, determine how the computer was infected.
- Based on the previous answer, what is the SHA256 hash for the file that probably infected the computer?
- The pcap contains 3 Windows executable files sent over HTTP. Export them from the pcap. What are the SHA256 file hashes of the those 3 files?

### ANSWERS:

MAC address: 00:15:c5:de:c7:3b (Dell\_de:c7:3b)

IP address: 192.168.1.96

Host name: FlashGordon-PC

Date/Time of the infection: 2017-06-27 at 13:38 UTC

FlashGordon-PC appears to be infected with a [Pushdo](#) malware variant

FlashGordon-PC was probably infected from Japanese malicious spam (malspam) matching characteristics reported at: <https://myonlinesecurity.co.uk/japanese-language-invoice-malspam-using-js-files-inside-zips-today/>

The report on this malspam noted zip attachments with extracted .js files having a SHA256 hashes of:

59727ea201d94d471bdbb353b1281cca3c5bebf5305b140a3c017e52a37be702  
64ae13592c5d8b289be81ebb50e054cef49b0bdf50ea92dd44ed611dd274150f

So it could be either one of those, or another file hash possibly related to them.

Three executables noted in the pcap:

gerv.gun - SHA256 hash:  
0931537889c35226d00ed26962ecacb140521394279eb2ade7e9d2afcf1a7272

wp.exe - SHA256 hash:  
79d503165d32176842fe386d96c04fb70f6ce1c8a485837957849297e625ea48

trow.exe - SHA256 hash:  
94a0a09ee6a21526ac34d41eabf4ba603e9a30c26e6a1dc072ff45749dfb1fe1

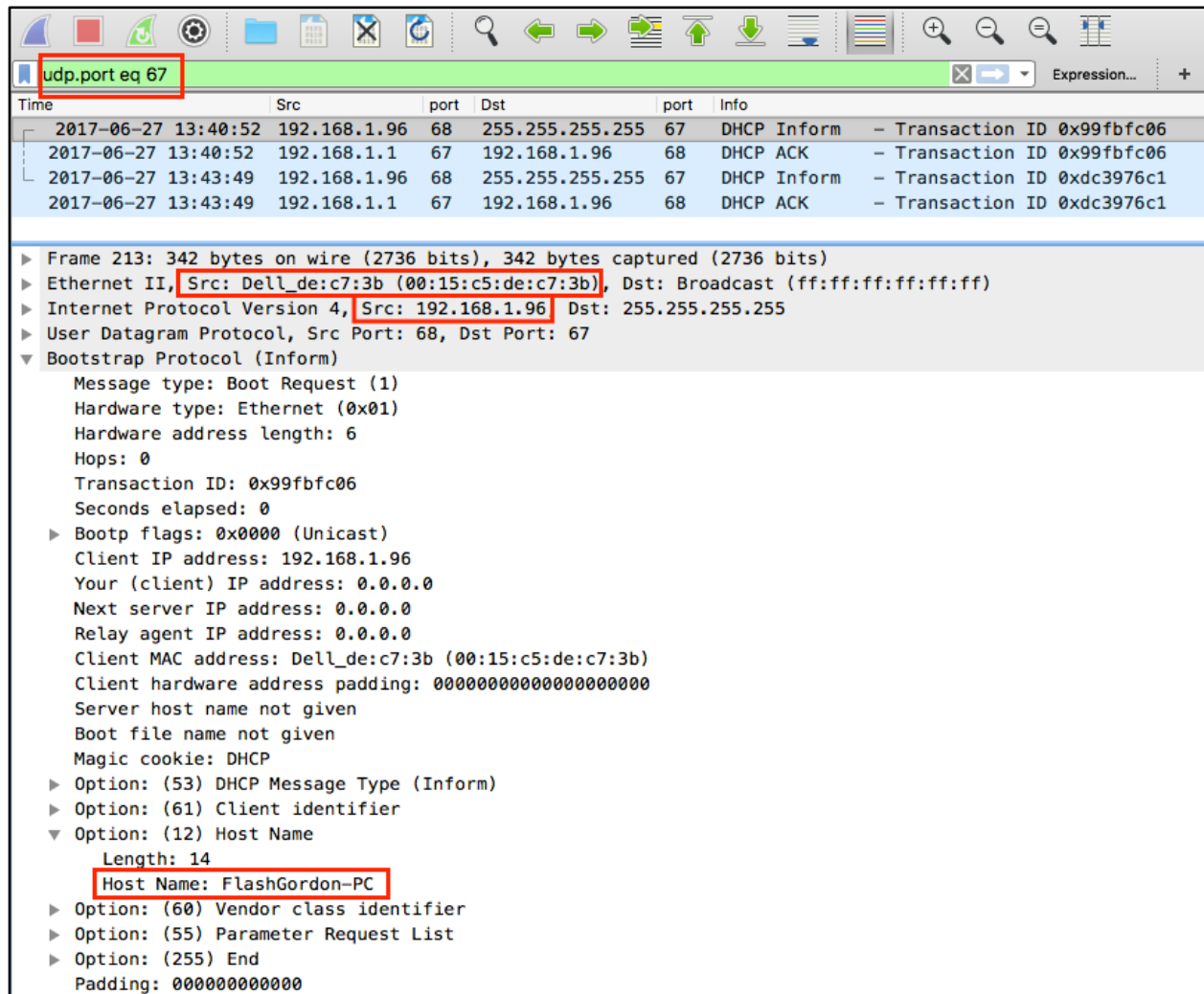
## 2017-06-28 TRAFFIC ANALYSIS EXERCISE - ANSWERS

### DETAILS:

As always, I recommend you set up Wireshark in accordance with my guide at:

- <http://www.malware-traffic-analysis.net/tutorials/wireshark/index.html>

The IP address is easy, because that was the IP address used to retrieve this traffic for the pcaps. Filter on DHCP traffic by using **udp.port eq 67** (or you can use **udp.port eq 68**), and you'll find the info as noted in the image below.



Shown above: DHCP traffic in the pcap, and where to find the host info.

Date and time of the infection is easy enough, since it's the first HTTP GET request. All HTTP traffic in this pcap is related to the infection.

If you look at the Snort and Suricata alerts, you'll find many of them are some sort of alert for Pusdho malware, so this is likely a Pushdo infection.

## 2017-06-28 TRAFFIC ANALYSIS EXERCISE - ANSWERS

| CNT | Date/Time     | Src IP         | SPort | Dst IP          | DPort | Pr | Event Message                                        |
|-----|---------------|----------------|-------|-----------------|-------|----|------------------------------------------------------|
| 1   | 2017-06-27... | 119.28.70.207  | 80    | 192.168.1.96    | 49184 | 6  | ET POLICY PE EXE or DLL Windows file download        |
| 1   | 2017-06-27... | 119.28.70.207  | 80    | 192.168.1.96    | 49184 | 6  | ET POLICY Binary Download Smaller than 1 MB Li       |
| 1   | 2017-06-27... | 119.28.70.207  | 80    | 192.168.1.96    | 49184 | 6  | ET CURRENT_EVENTS WinHttpRequest Downloadin          |
| 2   | 2017-06-27... | 192.168.1.96   | 49190 | 145.131.10.21   | 80    | 6  | ET POLICY exe download via HTTP - Informationa       |
| 4   | 2017-06-27... | 145.131.10.21  | 80    | 192.168.1.96    | 49190 | 6  | ET POLICY PE EXE or DLL Windows file download        |
| 4   | 2017-06-27... | 145.131.10.21  | 80    | 192.168.1.96    | 49190 | 6  | ET POLICY Binary Download Smaller than 1 MB Li       |
| 823 | 2017-06-27... | 192.168.1.96   | 49200 | 96.82.200.1     | 80    | 6  | ET TROJAN Backdoor.Win32.Pushdo.s Checkin            |
| 1   | 2017-06-27... | 192.168.1.96   | 49200 | 96.82.200.1     | 80    | 6  | ET POLICY Internet Explorer 6 in use - Significant S |
| 2   | 2017-06-27... | 144.217.254.90 | 80    | 192.168.1.96    | 49227 | 6  | ETPRO TROJAN Sinkhole.tech Sinkhole Reply            |
| 1   | 2017-06-27... | 192.168.1.96   | 49266 | 193.166.255.... | 80    | 6  | ET TROJAN Connection to Fitsec Sinkhole IP (Possi    |
| 1   | 2017-06-27... | 212.61.180.100 | 80    | 192.168.1.96    | 49533 | 6  | ET TROJAN Possible Compromised Host Sinkhole         |

Shown above: Summary of the Suricata alerts grouped according to Source IP.

```

2017-06-28-traffic-analysis-exercise-Snort-alerts.txt (~/.Downloads) - gedit
File Edit View Search Tools Documents Help
2017-06-28-traffic-...ise-Snort-alerts.txt x
[Classification: A Network Trojan was detected] [Priority: 1]
06/27-13:44:14.113561 192.168.1.96:49582 -> 157.7.222.253:80
TCP TTL:47 TOS:0x0 ID:22740 IpLen:20 DgmLen:670 DF
***A**** Seq: 0x4474C5A6 Ack: 0xEA758DB2 Win: 0x1BBC TcpLen: 20

[**] [1:29891:7] MALWARE-CNC Win.Trojan.Pushdo variant outbound connection [**]
[Classification: A Network Trojan was detected] [Priority: 1]
06/27-13:44:14.262646 192.168.1.96:49601 -> 209.85.25.218:80
TCP TTL:50 TOS:0x0 ID:54904 IpLen:20 DgmLen:1386 DF
***A**** Seq: 0xB81902C2 Ack: 0x57C05764 Win: 0x2390 TcpLen: 20

[**] [1:29891:7] MALWARE-CNC Win.Trojan.Pushdo variant outbound connection [**]
[Classification: A Network Trojan was detected] [Priority: 1]
06/27-13:44:14.256556 192.168.1.96:49599 -> 173.203.43.103:80
TCP TTL:110 TOS:0x0 ID:17822 IpLen:20 DgmLen:796 DF
***AP**F Seq: 0xB5B6B29F Ack: 0xD1BC22EB Win: 0x200 TcpLen: 20

[**] [1:29891:7] MALWARE-CNC Win.Trojan.Pushdo variant outbound connection [**]
[Classification: A Network Trojan was detected] [Priority: 1]
06/27-13:44:14.166546 192.168.1.96:49588 -> 136.243.9.216:80
TCP TTL:43 TOS:0x0 ID:30184 IpLen:20 DgmLen:1902 DF
***A**** Seq: 0xFBFB4921 Ack: 0x6DA41B03 Win: 0x4900 TcpLen: 20

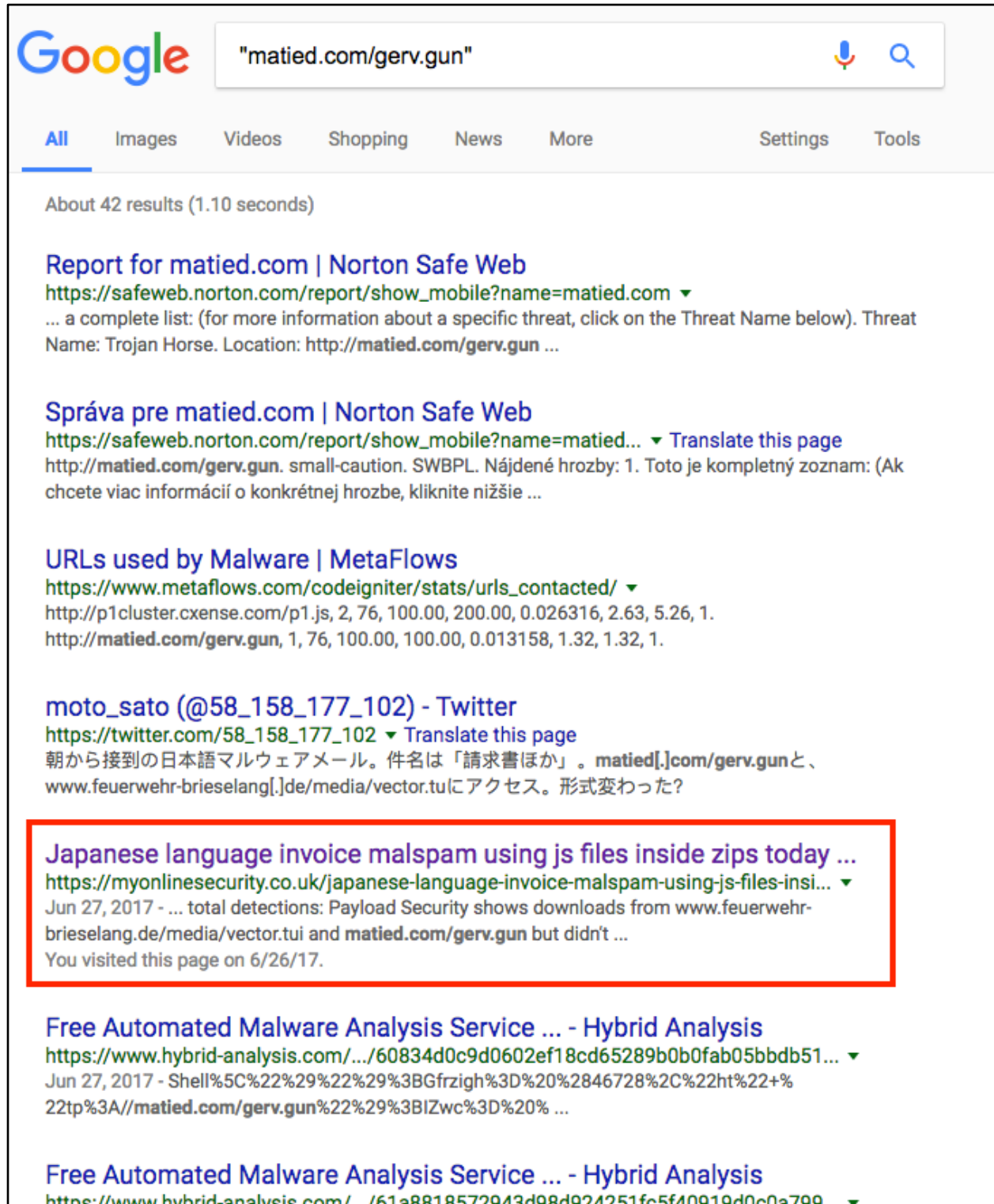
[**] [1:10887:7] INDICATOR_OBFUSCATION_potential_suspicious_unescape_obfuscation

```

Shown above: Some of the many Pushdo alerts in the file with all the Snort alerts/.

How did this infection happen? One of the quickest ways to find information is to look at the related HTTP requests from the infection traffic. Doing a Google search on the first HTTP request will return a variety of results. Remember to use quotation marks around the URL in the Google search bar, or your browser might try viewing the URL. Look for any articles written about how the malware may have been delivered. In this case, you should find one from myonlinesecurity.co.uk discussing Japanese malspam using .js files inside zip archives attached to the emails.

## 2017-06-28 TRAFFIC ANALYSIS EXERCISE - ANSWERS



The screenshot shows a Google search interface with the query "matied.com/gerv.gun". The search results are displayed below the navigation bar. The first result is a report from Norton Safe Web, followed by a similar report in Japanese. The third result is from MetaFlows, showing URLs used by malware. The fourth result is a tweet from @58\_158\_177\_102. The fifth result, highlighted with a red box, is an article from myonlinesecurity.co.uk about Japanese language invoice malware. The sixth and seventh results are from Hybrid Analysis, showing automated malware analysis services.

Google

"matied.com/gerv.gun"

All Images Videos Shopping News More Settings Tools

About 42 results (1.10 seconds)

**Report for matied.com | Norton Safe Web**  
[https://safeweb.norton.com/report/show\\_mobile?name=matied.com](https://safeweb.norton.com/report/show_mobile?name=matied.com) ▼  
... a complete list: (for more information about a specific threat, click on the Threat Name below). Threat Name: Trojan Horse. Location: <http://matied.com/gerv.gun> ...

**Správa pre matied.com | Norton Safe Web**  
[https://safeweb.norton.com/report/show\\_mobile?name=matied...](https://safeweb.norton.com/report/show_mobile?name=matied...) ▼ [Translate this page](#)  
<http://matied.com/gerv.gun>. small-caution. SWBPL. Nájdené hrozby: 1. Toto je kompletný zoznam: (Ak chcete viac informácií o konkrétnej hrozbe, kliknite nižšie ...

**URLs used by Malware | MetaFlows**  
[https://www.metaflows.com/codeigniter/stats/urls\\_contacted/](https://www.metaflows.com/codeigniter/stats/urls_contacted/) ▼  
<http://p1cluster.cxense.com/p1.js>, 2, 76, 100.00, 200.00, 0.026316, 2.63, 5.26, 1.  
<http://matied.com/gerv.gun>, 1, 76, 100.00, 100.00, 0.013158, 1.32, 1.32, 1.

**moto\_sato (@58\_158\_177\_102) - Twitter**  
[https://twitter.com/58\\_158\\_177\\_102](https://twitter.com/58_158_177_102) ▼ [Translate this page](#)  
朝から接到の日本語マルウェアメール。件名は「請求書ほか」。matied[.]com/gerv.gunと、www.feuerwehr-brieselang[.]de/media/vector.tuにアクセス。形式変わった？

**Japanese language invoice malware using js files inside zips today ...**  
<https://myonlinesecurity.co.uk/japanese-language-invoice-malware-using-js-files-insi...> ▼  
Jun 27, 2017 - ... total detections: Payload Security shows downloads from www.feuerwehr-brieselang.de/media/vector.tui and [matied.com/gerv.gun](http://matied.com/gerv.gun) but didn't ...  
You visited this page on 6/26/17.

**Free Automated Malware Analysis Service ... - Hybrid Analysis**  
<https://www.hybrid-analysis.com/.../60834d0c9d0602ef18cd65289b0b0fab05bbdb51...> ▼  
Jun 27, 2017 - Shell%5C%22%29%22%29%3BGfrzigh%3D%20%2846728%2C%22ht%22+%22tp%3A//matied.com/gerv.gun%22%29%3BIZwc%3D%20% ...

**Free Automated Malware Analysis Service ... - Hybrid Analysis**  
<https://www.hybrid-analysis.com/.../61a8818572943d98d924251fc5f40919d0c0a799> ▼

Shown above: Google search results, and an article about malware.

The blog post from myonlinesecurity.co.uk contains links to VirusTotal entries for the associated files.

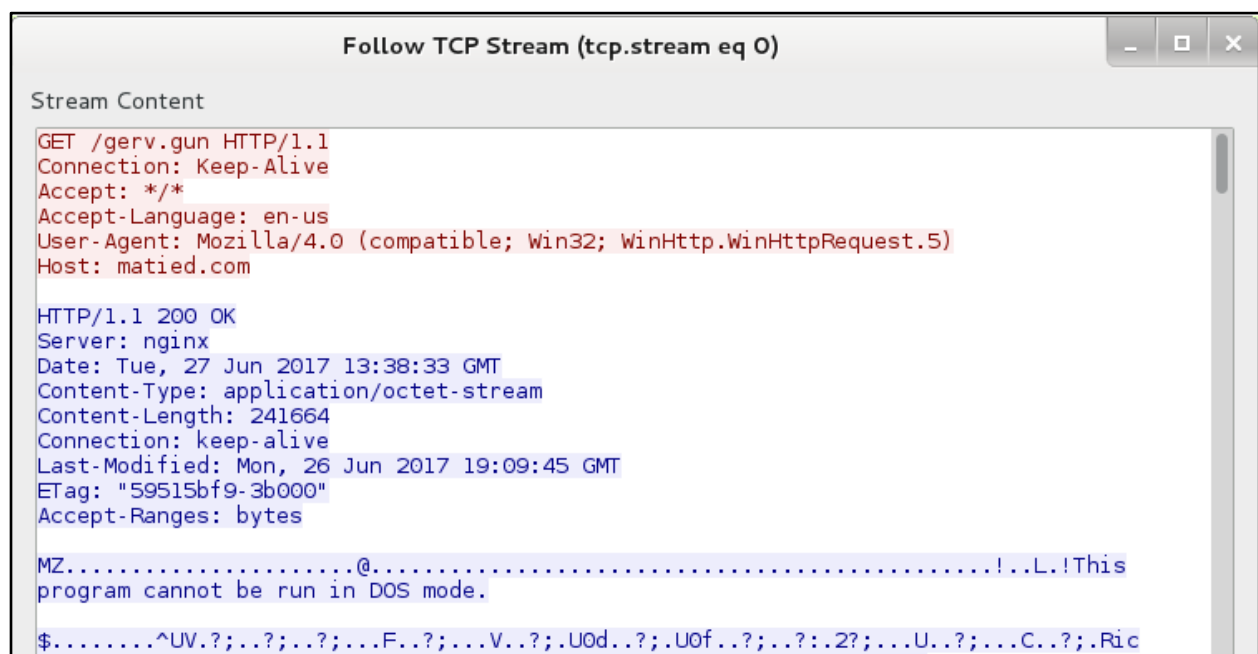
## 2017-06-28 TRAFFIC ANALYSIS EXERCISE - ANSWERS



Shown above: Links from the [myonlinesecurity.co.uk](http://myonlinesecurity.co.uk) article for the .js file info.

You can check to see if the Payload Security links have any other HTTP requests in the sandbox analysis that match traffic from the exercise pcap.

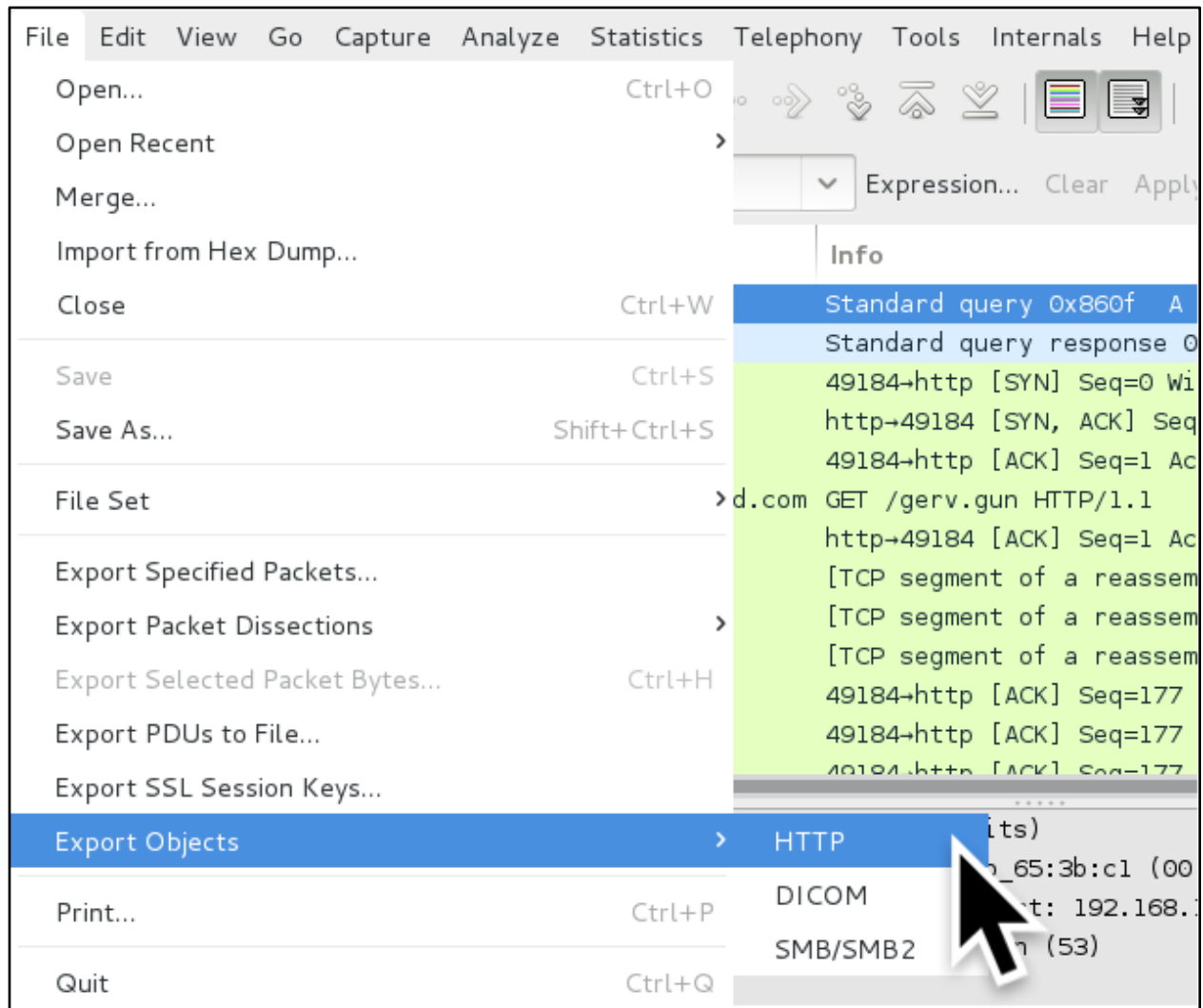
If you follow the first TCP stream in the pcap, you'll find the HTTP GET request for gerv.gun returns an executable file.



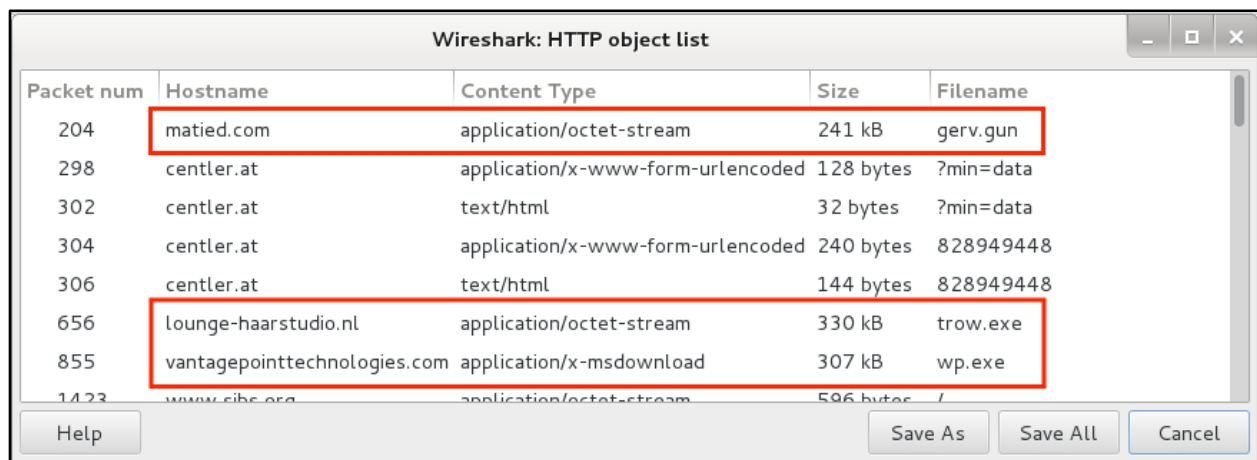
Shown above: TCP stream for the first HTTP request in our exercise pcap.

## 2017-06-28 TRAFFIC ANALYSIS EXERCISE - ANSWERS

To extract this and other executables from the pcap, go to the menu for **File → Export Object → HTTP**



Shown above: Exporting HTTP objects from the pcap.



Shown above: Three executables you can extract from the pcap.

## 2017-06-28 TRAFFIC ANALYSIS EXERCISE - ANSWERS

---

From the Wireshark HTTP object list, in addition to gerv.gun, you can see entries for throw.exe and wp.exe. Check the TCP streams for the associated HTTP requests for the files to make sure they are indeed executable files. Select each one from the HTTP object list and click "Save As"

A quick check on a Linux system with the "file" command shows these are executable files.

```
# file *.exe *.gun
throw.exe: PE32 executable (GUI) Intel 80386, for MS Windows
wp.exe:    PE32 executable (GUI) Intel 80386, for MS Windows
gerv.gun:  PE32 executable (GUI) Intel 80386, for MS Windows
# █
```

*Shown above: Using the "file" command on the extracted files.*

The "shasum" command will give you the SHA256 file hashes for the extracted files as shown below, or you can submit them to VirusTotal and get the information from the analysis results.

```
# shasum -a 256 *.exe *.gun
94a0a09ee6a21526ac34d41eabf4ba603e9a30c26e6a1dc072ff45749dfb1fe1  throw.exe
79d503165d32176842fe386d96c04fb70f6ce1c8a485837957849297e625ea48  wp.exe
0931537889c35226d00ed26962ecacb140521394279eb2ade7e9d2afc1a7272  gerv.gun
# █
```

*Shown above: Using the "shasum" command to find the SHA256 hashes of the files.*

There's certainly more you can investigate and document for this infection; however, we've now covered all the objectives for this exercise, and you can write a good incident report based on the information we've uncovered.